

|                          |                                                                                 |                  |            |
|--------------------------|---------------------------------------------------------------------------------|------------------|------------|
| <b>Class:</b>            | CPE417 / ECG617                                                                 | <b>Semester:</b> | Fall 2024  |
| <b>Assignment type:</b>  | Tutorial                                                                        | <b>Due date:</b> | See canvas |
| <b>Assignment topic:</b> | Arduino MQTT setup                                                              |                  |            |
| <b>Delivery:</b>         | WebCampus<br>Submit zip file with all the files listed in the "Submission" part |                  |            |

I first compiled and ran the code above to test the temperature sensor

```
sketch_nov19a.ino
1  #include <WiFi101.h>
2
3  #define tempPin  A1
4  #define buttonPin  A0
5  #define photoPin  A4
6  float v = 0.0;
7
8  void setup() {
9      // put your setup code here, to run once:
10     Serial.begin(9600);
11     WiFi.begin();
12     Serial.print("RRSI: ");
13     Serial.println(WiFi.RSSI());
14 }
15
16 void loop() {
17     // put your main code here, to run repeatedly:
18     int buttonState = digitalRead(buttonPin);
19     if(buttonState == LOW)
```

Output Serial Monitor X

Message (Enter to send message to 'Arduino MKR 1000 WIFI' on 'COM7')

```
19:55:38.229 -> temperature : -90.09 F
19:55:38.229 -> temperature : -90.09 F
19:55:38.229 -> temperature : -90.09 F
19:55:38.229 -> temperature : -89.59 F
19:55:38.229 -> temperature : -90.09 F
19:55:38.229 -> temperature : -89.59 F
19:55:38.229 -> temperature : -89.59 F
19:55:38.229 -> temperature : -90.09 F
19:55:38.229 -> temperature : -89.59 F
```

```
sketch_nov19a.ino
1  #include <WiFi101.h>
2
3  #define tempPin  A1
4  #define buttonPin  A0
5  #define photoPin  A4
6  float v = 0.0;
7
8  void setup() {
9      // put your setup code here, to run once:
10     Serial.begin(9600);
11     WiFi.begin();
12     Serial.print("RRSI: ");
13     Serial.println(WiFi.RSSI());
14 }
15
16 void loop() {
17     // put your main code here, to run repeatedly:
18     int buttonState = digitalRead(buttonPin);
19     if(buttonState == LOW)
20     {
21         float heat = analogRead(tempPin);
22         v = heat * (5000/1024);
23     }
24     float realHeat = 25-((0.75+v)/(10^2));
25     Serial.print("temperature : ");
26     Serial.print(realHeat);
27     Serial.print(" F\n");
28 }
```

Next, I downloaded the mosquitto software and ran it on both my laptop and computer.

## Source

- [mosquitto-2.0.20.tar.gz](#) (GPG signature)
- [Git source code repository](#) (github.com)

Older downloads are available at <https://mosquitto.org/files/>

## Binary Installation

The binary packages listed below are supported by the Mosquitto project. In many cases Mosquitto is also available directly from official Linux/BSD distributions.

## Windows

- 64-bit: [mosquitto-2.0.20-install-windows-x64.exe](#)
- 32-bit: [mosquitto-2.0.20-install-windows-x86.exe](#)

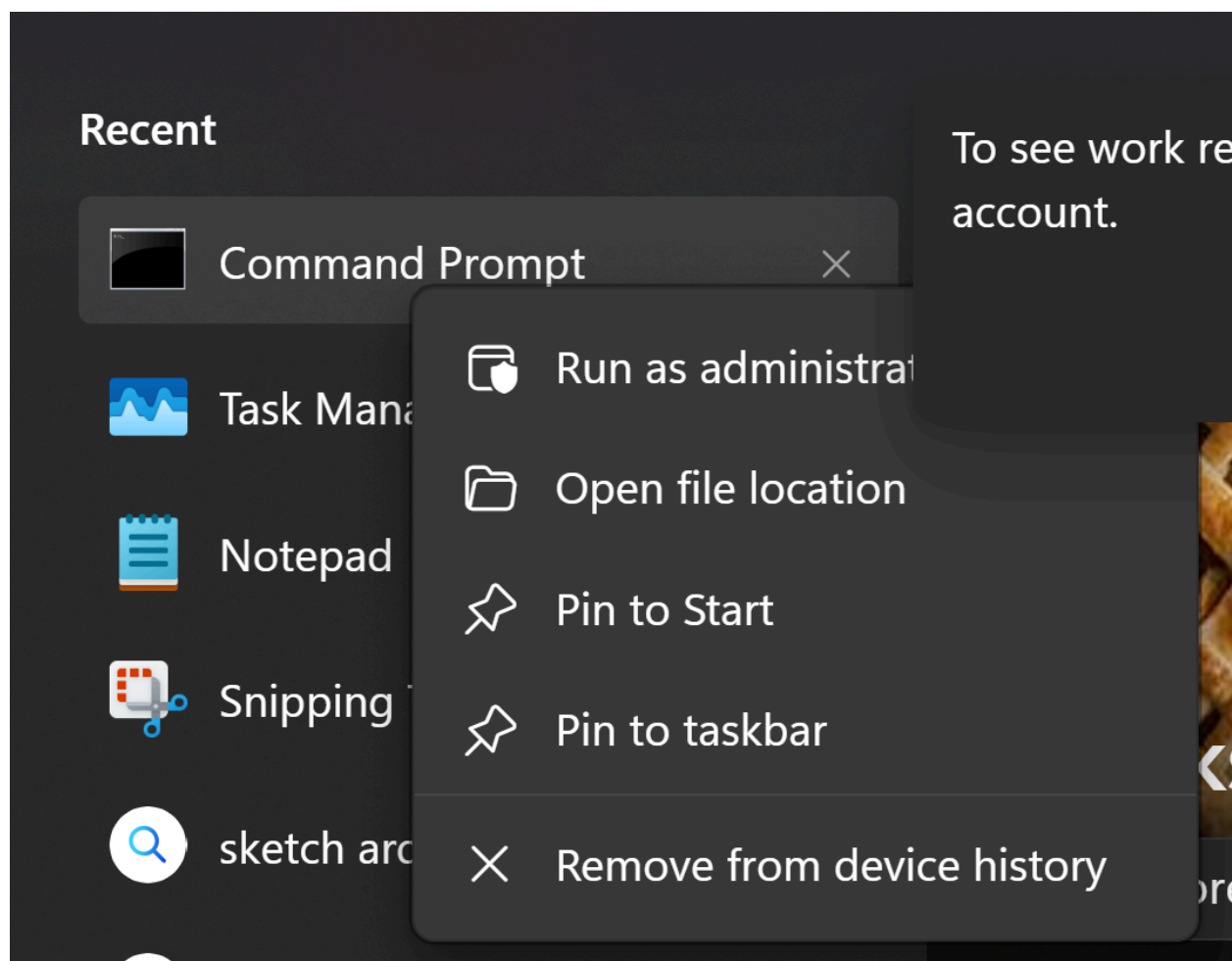
Older installers can be found at <https://mosquitto.org/files/binary/>.

See also README-windows.md after installing.

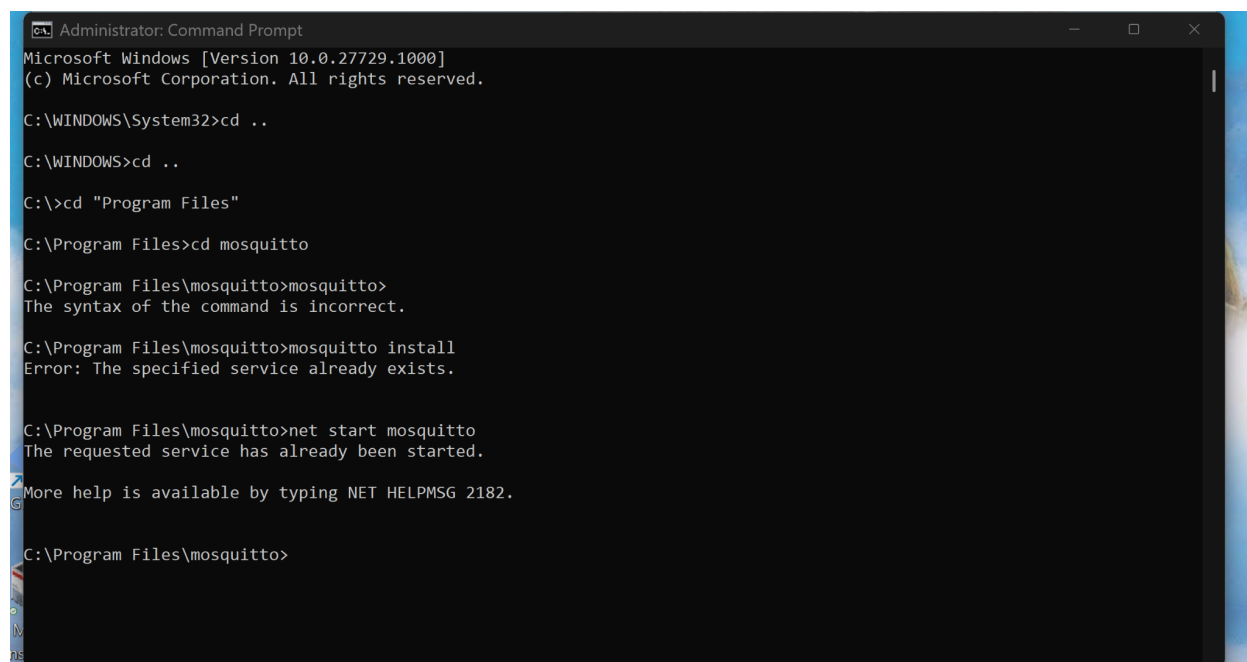
## Mac

Mosquitto can be installed from the homebrew project. See [brew.sh](#) and then use `brew install mosquitto`





Once the download was finished, I opened the command prompt as an administrator.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.27729.1000]
(c) Microsoft Corporation. All rights reserved.

C:\WINDOWS\System32>cd ..
C:\WINDOWS>cd ..
C:\>cd "Program Files"
C:\Program Files>cd mosquitto
C:\Program Files\mosquitto>mosquitto>
The syntax of the command is incorrect.

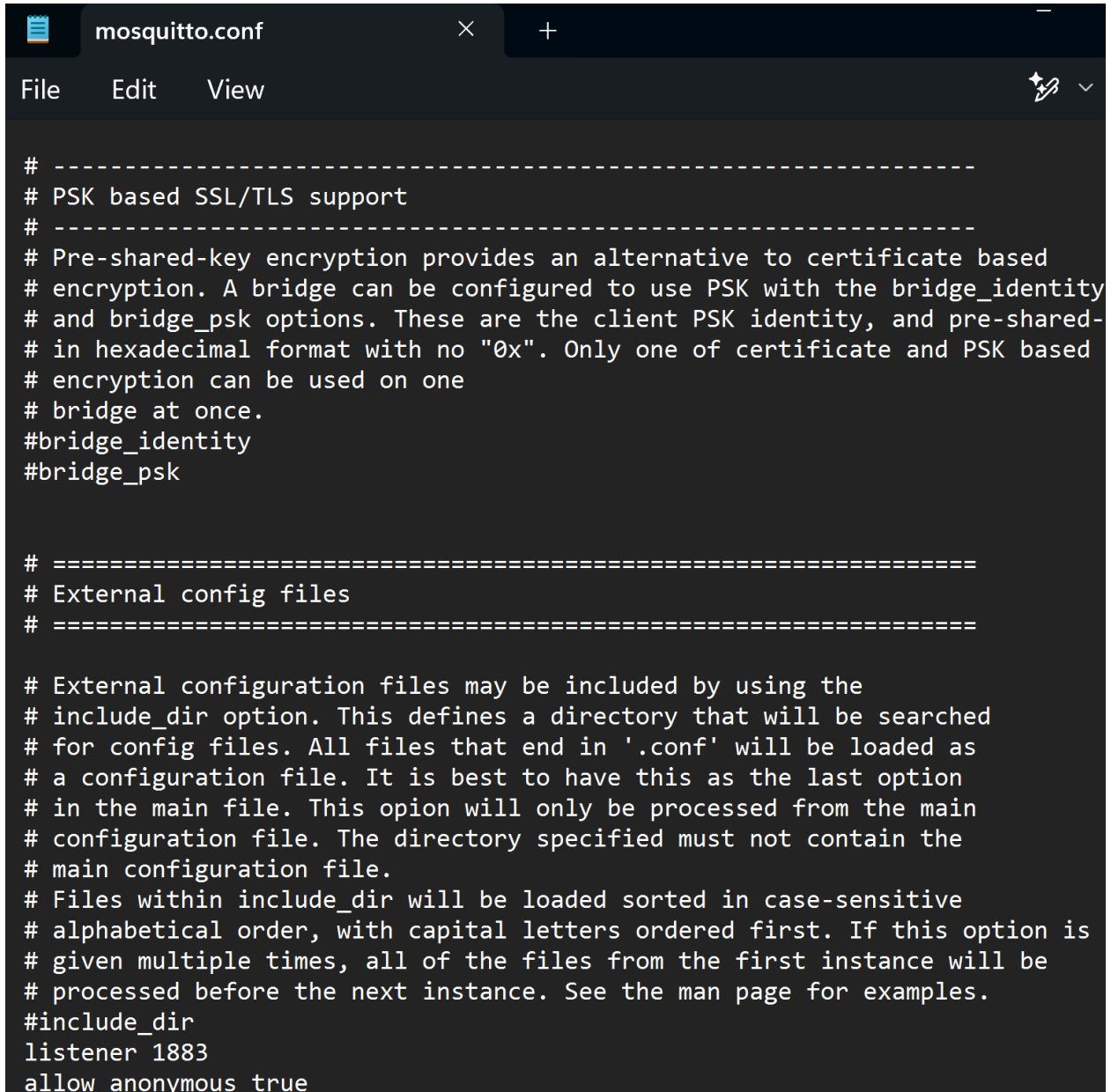
C:\Program Files\mosquitto>mosquitto install
Error: The specified service already exists.

C:\Program Files\mosquitto>net start mosquitto
The requested service has already been started.

More help is available by typing NET HELPMSG 2182.

C:\Program Files\mosquitto>
```

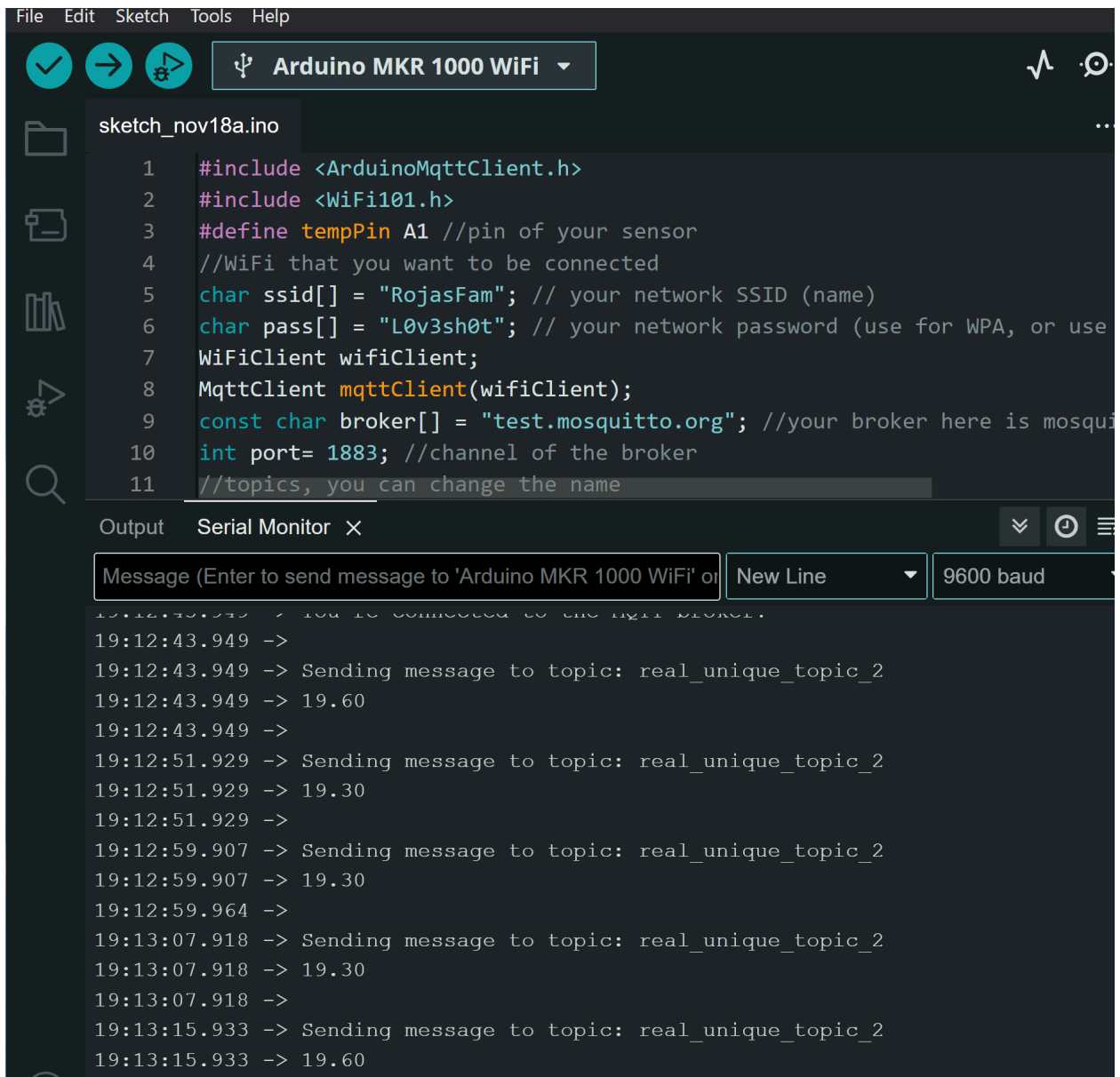
Then I changed directories to locate where my mosquitto program was. Then I type mosquitto install to verify it has installed correctly. After that I type net start mosquitto. I do this for both my client program which is on my computer and publisher on my laptop.



```
# -----  
# PSK based SSL/TLS support  
# -----  
# Pre-shared-key encryption provides an alternative to certificate based  
# encryption. A bridge can be configured to use PSK with the bridge_identity  
# and bridge_psk options. These are the client PSK identity, and pre-shared-  
# in hexadecimal format with no "0x". Only one of certificate and PSK based  
# encryption can be used on one  
# bridge at once.  
#bridge_identity  
#bridge_psk  
  
# =====  
# External config files  
# =====  
  
# External configuration files may be included by using the  
# include_dir option. This defines a directory that will be searched  
# for config files. All files that end in '.conf' will be loaded as  
# a configuration file. It is best to have this as the last option  
# in the main file. This option will only be processed from the main  
# configuration file. The directory specified must not contain the  
# main configuration file.  
# Files within include_dir will be loaded sorted in case-sensitive  
# alphabetical order, with capital letters ordered first. If this option is  
# given multiple times, all of the files from the first instance will be  
# processed before the next instance. See the man page for examples.  
#include_dir  
listener 1883  
allow_anonymous true
```

During this period, I also changed the configuration file to include port 1883 and allow\_anonymous true in order to be able to use my ip address for the programs to work.

Publisher:



```
File Edit Sketch Tools Help

✓ → ⚙️ Arduino MKR 1000 WiFi

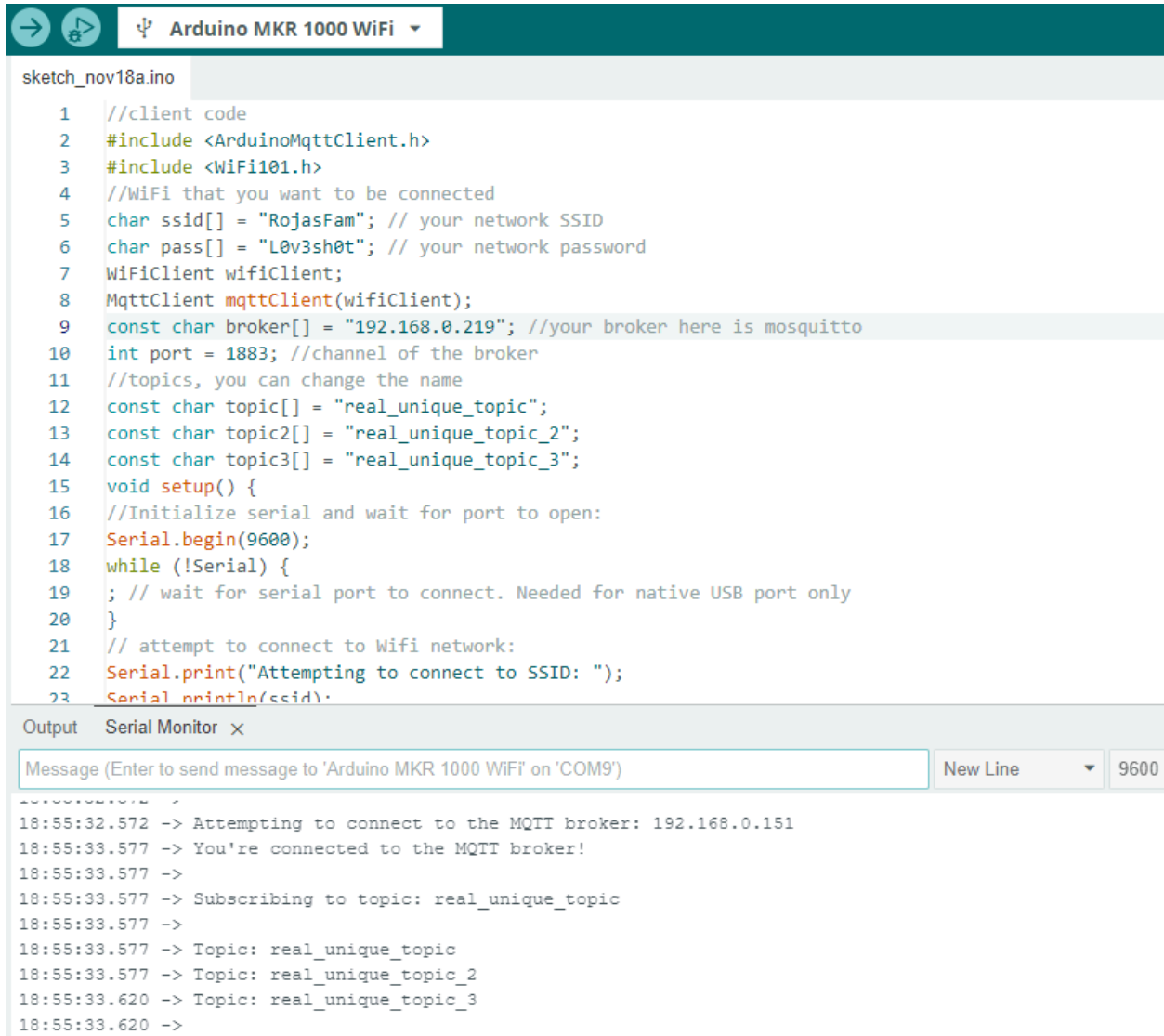
sketch_nov18a.ino

1 #include <ArduinoMqttClient.h>
2 #include <WiFi101.h>
3 #define tempPin A1 //pin of your sensor
4 //WiFi that you want to be connected
5 char ssid[] = "RojasFam"; // your network SSID (name)
6 char pass[] = "L0v3sh0t"; // your network password (use for WPA, or use
7 WiFiClient wifiClient;
8 MqttClient mqttClient(wifiClient);
9 const char broker[] = "test.mosquitto.org"; //your broker here is mosquit
10 int port= 1883; //channel of the broker
11 //topics, you can change the name

Output Serial Monitor X
Message (Enter to send message to 'Arduino MKR 1000 WiFi' or) New Line 9600 baud

19:12:43.949 -> You are connected to the MQTT broker.
19:12:43.949 ->
19:12:43.949 -> Sending message to topic: real_unique_topic_2
19:12:43.949 -> 19.60
19:12:43.949 ->
19:12:51.929 -> Sending message to topic: real_unique_topic_2
19:12:51.929 -> 19.30
19:12:51.929 ->
19:12:59.907 -> Sending message to topic: real_unique_topic_2
19:12:59.907 -> 19.30
19:12:59.964 ->
19:13:07.918 -> Sending message to topic: real_unique_topic_2
19:13:07.918 -> 19.30
19:13:07.918 ->
19:13:15.933 -> Sending message to topic: real_unique_topic_2
19:13:15.933 -> 19.60
```

Then I compile and run the publisher program.



```
sketch_nov18a.ino
1 //client code
2 #include <ArduinoMqttClient.h>
3 #include <WiFi101.h>
4 //WiFi that you want to be connected
5 char ssid[] = "RojasFam"; // your network SSID
6 char pass[] = "L0v3sh0t"; // your network password
7 WiFiClient wifiClient;
8 MqttClient mqttClient(wifiClient);
9 const char broker[] = "192.168.0.219"; //your broker here is mosquitto
10 int port = 1883; //channel of the broker
11 //topics, you can change the name
12 const char topic[] = "real_unique_topic";
13 const char topic2[] = "real_unique_topic_2";
14 const char topic3[] = "real_unique_topic_3";
15 void setup() {
16 //Initialize serial and wait for port to open:
17 Serial.begin(9600);
18 while (!Serial) {
19 ; // wait for serial port to connect. Needed for native USB port only
20 }
21 // attempt to connect to Wifi network:
22 Serial.print("Attempting to connect to SSID: ");
23 Serial.println(ssid);
24 }

Output Serial Monitor x
Message (Enter to send message to 'Arduino MKR 1000 WiFi' on 'COM9') New Line 9600

18:55:32.572 -> Attempting to connect to the MQTT broker: 192.168.0.151
18:55:33.577 -> You're connected to the MQTT broker!
18:55:33.577 ->
18:55:33.577 -> Subscribing to topic: real_unique_topic
18:55:33.577 ->
18:55:33.577 -> Topic: real_unique_topic
18:55:33.577 -> Topic: real_unique_topic_2
18:55:33.620 -> Topic: real_unique_topic_3
18:55:33.620 ->
```

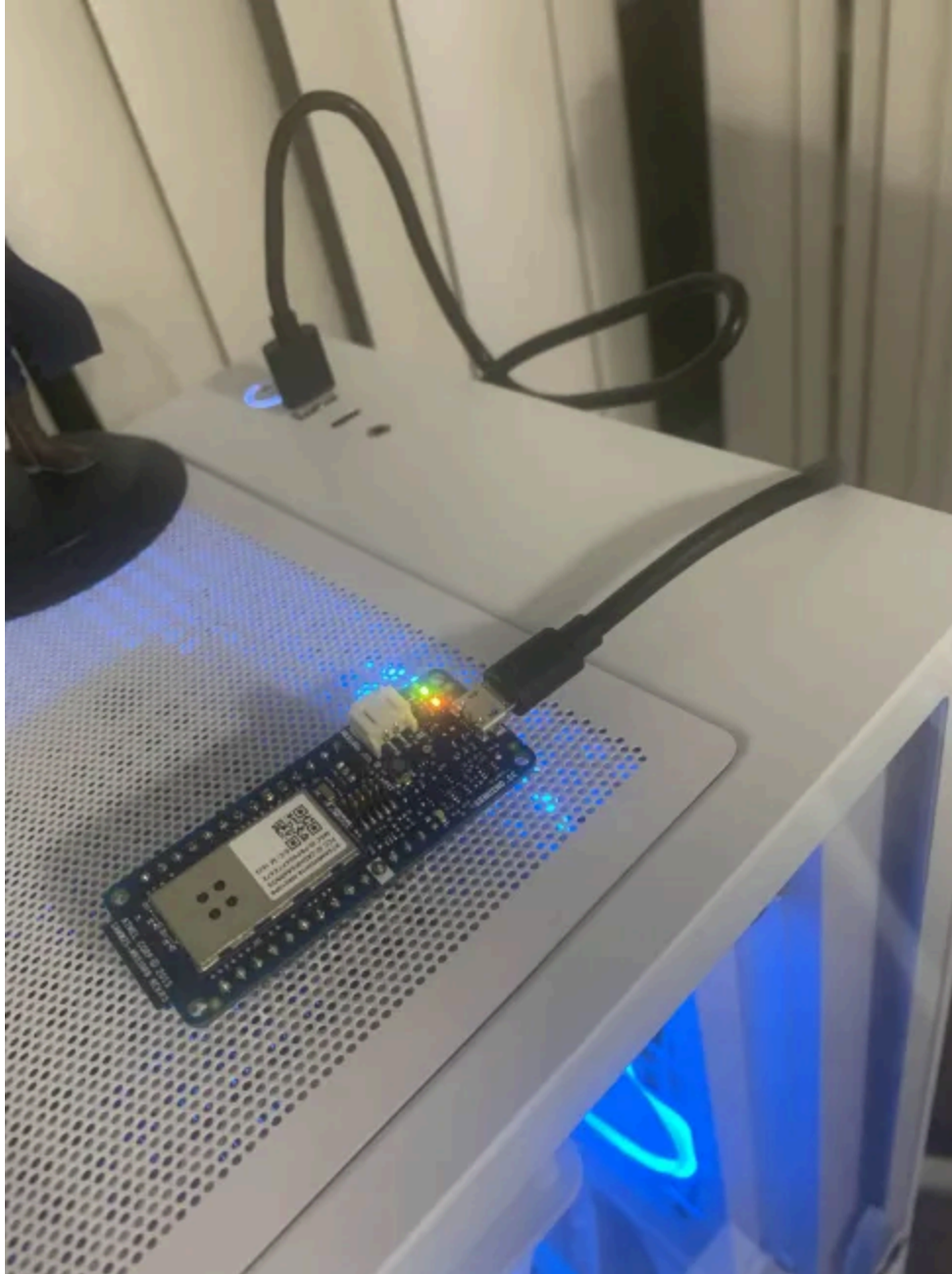
```
sketch_nov18a.ino
5 //WiFi that you want to be connected
6 char ssid[] = "RojasFam"; // your network SSID (name)
7 char pass[] = "L0v3sh0t"; // your network password (use for WPA, or use
8 WiFiClient wifiClient;
9 MqttClient mqttClient(wifiClient);
10 const char broker[] = "192.168.0.219"; //your broker here is mosquitto
11 int port= 1883; //channel of the broker
12 //topics, you can change the name
13 const char topic[] = "real_unique_topic";
14 const char topic2[] = "real_unique_topic_2";
15 const char topic3[] = "real_unique_topic_3";
16 //set interval for sending messages (milliseconds)
17 const long interval = 8000;
```

Output Serial Monitor X

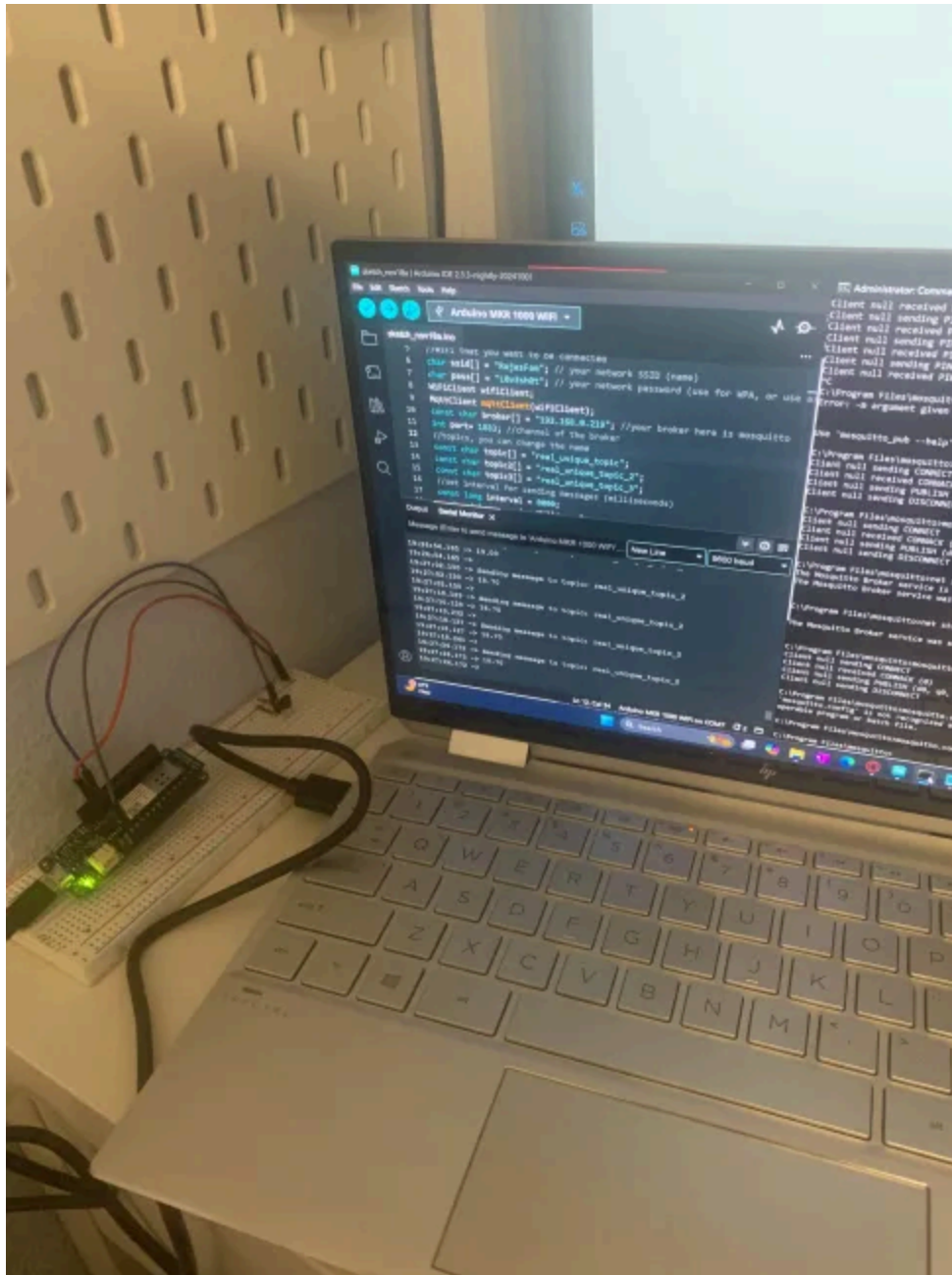
Message (Enter to send message to 'Arduino MKR 1000 WiFi'... New Line 9600 baud

```
19:23:42.165 -> 18.40
19:23:42.165 ->
19:23:50.160 -> Sending message to topic: real_unique_topic_2
19:23:50.160 -> 18.70
19:23:50.160 ->
19:23:58.141 -> Sending message to topic: real unique topic 2
```

Next I ran and compiled my client program on my computer and the publisher program on my laptop. I used 2 arduino mkr1000 boards. One is connected to my laptop along with the temperature sensor and the other is connected to my computer.



This was the connection from arduino to my computer running the client program.



This was my setup from temperature sensor to arduino to laptop for my publisher program.

```
C:\Program Files\mosquitto>mosquitto_sub -t real_unique_topic_2/# -v -h 192.168.0.219
real_unique_topic_2 18.40
real_unique_topic_2 18.40
real_unique_topic_2 18.10
real_unique_topic_2 18.10
real_unique_topic_2 18.10
real_unique_topic_2 18.10
real_unique_topic_2 18.10
real_unique_topic_2 18.70
real_unique_topic_2 18.40
real_unique_topic_2 17.80
real_unique_topic_2 17.80
real_unique_topic_2 17.80
```

Next, I picked a topic from the ones listed in the sketch terminal. Automatically, it started printing the temperature results from the sensor.

Client:

```
19:21:51.199 ->
19:21:59.227 -> Received a message with topic '
19:21:59.227 -> real_unique_topic_2', length 5 bytes:
19:21:59.227 -> 18.40
19:21:59.227 ->
19:22:07.213 -> Received a message with topic '
19:22:07.213 -> real_unique_topic_2', length 5 bytes:
19:22:07.213 -> 18.40
19:22:07.213 ->
19:22:15.216 -> Received a message with topic '
19:22:15.216 -> real_unique_topic_2', length 5 bytes:
19:22:15.216 -> 18.70
19:22:15.216 ->
19:22:23.205 -> Received a message with topic '
19:22:23.205 -> real_unique_topic_2', length 5 bytes:
19:22:23.205 -> 18.70
19:22:23.205 ->
19:22:31.219 -> Received a message with topic '
19:22:31.219 -> real_unique_topic_2', length 5 bytes:
19:22:31.219 -> 18.70
19:22:31.219 ->

real_unique_topic_2 18.40
real_unique_topic_2 18.70
real_unique_topic_2 18.40
real_unique_topic_2 18.40
real_unique_topic_2 18.70
real_unique_topic_2 18.70
real_unique_topic_2 18.70
real_unique_topic_2 18.70
real_unique_topic_2 18.40
real_unique_topic_2 18.40
real_unique_topic_2 18.10
real_unique_topic_2 18.40
real_unique_topic_2 18.10
real_unique_topic_2 18.70
real_unique_topic_2 18.70
real_unique_topic_2 18.70
real_unique_topic_2 18.70
real_unique_topic_2 18.40
real_unique_topic_2 19.00
real_unique_topic_2 18.40
real_unique_topic_2 18.40
real_unique_topic_2 18.40
real_unique_topic_2 18.70
real_unique_topic_2 18.70
real_unique_topic_2 18.70
```

These results are the results printed side by side of the sketch terminal and the mosquito terminal from the client program.

```
19:22:38.191 -> 18.70
19:22:38.191 ->
19:22:46.161 -> Sending message to topic: real_unique_topic_2
19:22:46.161 -> 18.70
19:22:46.161 ->
19:22:54.173 -> Sending message to topic: real_unique_topic_2
19:22:54.173 -> 19.00
19:22:54.173 ->
19:23:02.164 -> Sending message to topic: real_unique_topic_2
19:23:02.164 -> 18.70
19:23:02.164 ->
19:23:10.143 -> Sending message to topic: real_unique_topic_2
19:23:10.143 -> 18.70
19:23:10.244 ->
```

This is also the results from the main publisher program.

```
C:\Program Files\mosquitto>mosquitto_sub -h 192.168.0.219 -t real_unique_topic_2/#  
19.00
```

```
C:\Program Files\mosquitto>mosquitto_sub -h 192.168.0.219 -t real_unique_topic_2/# -C 3  
18.70  
18.70  
18.40
```

```
C:\Program Files\mosquitto>mosquitto_sub -t real_unique_topic_2/# -h 192.168.0.219  
18.40  
18.70  
18.40  
19.00  
18.40
```

Above I tried all the commands that were recommended in the pdf.